

# 2024 September Floods and Landslides

## Situation Report #4 as of 16 October 2024

(Third SitRep published on 03 October 2024)



Government of Nepal  
Ministry of Home Affairs  
National Disaster Risk Reduction and  
Management Authority

This report is produced by National Disaster Risk Reduction and Management Authority (NDRRMA) in collaboration with sectoral ministries, departments, provincial and local governments. It covers the situation of floods and landslides due to the heavy rainfall across the country from 26-28 September until 16 October 2024.

### Summary of Loss and Damage



**5,18,403**

Total HH affected



**2.59 Million**

Total Population affected



**46 Billion**

Estimated Economic Loss

#### Human Casualties



**250**  
Deaths



**76 Female**  
**114 Male**

2 Korean Citizens (M)  
1 American Citizen (M)



**60 Children**  
35 Boys, 25 Girls



**18**  
Missing



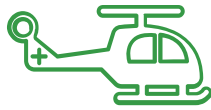
**178**  
Injured

#### Emergency Rescue



**30,731**

Security Personnel Deployed



**910**

Heli Rescued



**17,174**

Rescued

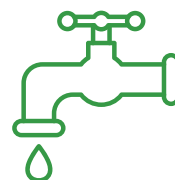
#### Water Supply and Sanitation

**1,639**

No of Projects worth  
NPR 5.73 Billion

**0.5 Million HHs**

**2.5 Million Population**



#### Private Housing



**5,916**

Fully Damaged



**12,477**

Partially Damaged



**10,807**

Displaced HHs



**18,403**

Private Housing needs

#### Infrastructure



**146 affected Bridges**  
NPR 1.04 Billion Loss



**7 Irrigation Projects**  
NPR 1.35 Billion Loss



**5 Fully Closed**  
**4 One-way Open**



**26 Hydropowers**  
**1,636.1 MW**  
NPR 3.02 Billion Loss

#### Telecommunications



**302 Projects**

NPR 151.8 Million  
(Nepal Telecom + NCELL)

#### Health and Education Sector



**Health**

52 affected



**Education**

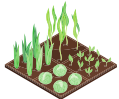
260 affected

#### Agriculture and Livelihood



**Paddy Field**

58,476 Hectares  
NPR 3.5 Billion Loss



**Fruits & Vegetables Farm**

4,765 Hectares



**Sugarcane Field**

85 Hectares  
NPR 0.34 Billion Loss



**Fish Pond**

458 Hectares  
NPR 1.13 Billion Loss



**Livestock**

26,205  
NPR 5.8 Billion

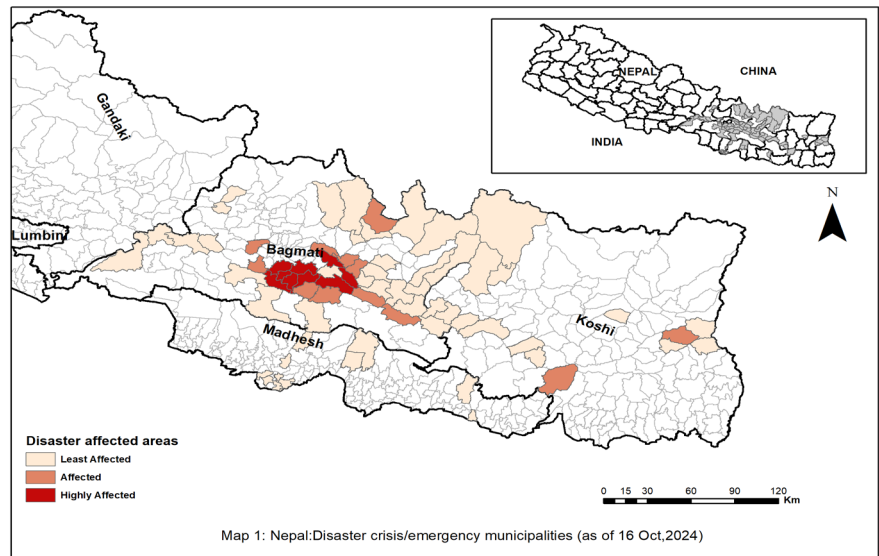


**Cardamom Field**

1,300 Hectares  
NPR 0.7 Billion Loss

## Highlights

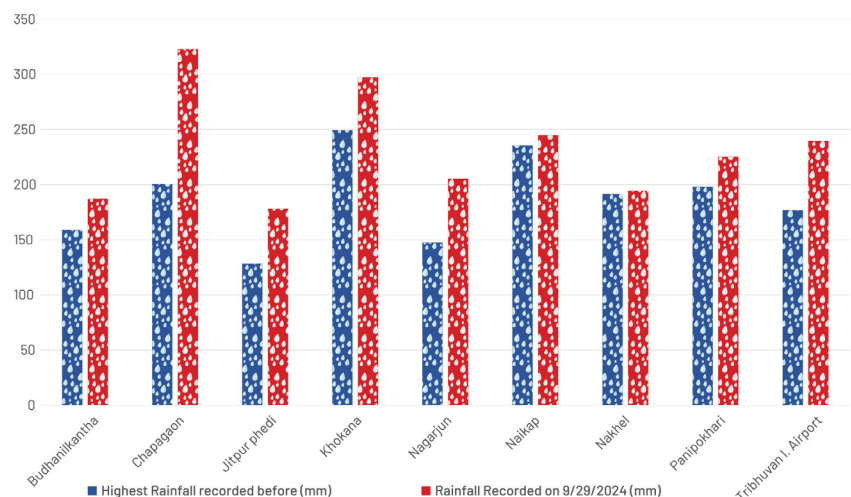
- 1) As of 16 October, the total number of deaths has reached **250** that includes 76 females, 114 males, and **60 children**, with **18 individuals still missing** and **178 reported injured**.
- 2) More than **17,000 people** have been **rescued** through various rescue operations that deployed more than 30 thousand security personnel.
- 3) The Nepal Government has declared **71 municipalities** across 20 districts as **disaster crisis/emergency zones** due to the severe impacts of landslides and floods.
- 4) **Bagmati Province** has distributed **NPR 58 Million** from the Provincial Disaster Management Fund to support **45 disaster crisis/emergency municipalities**.



## Situation Overview

Unprecedented rainfall during 27-29 September 2024 resulted in multiple incidents of flood and inundation along with small, medium and large scale landslides across the country. According to the Department of Hydrology and Meteorology (DHM), out of the 222 measurement stations where rainfall was simultaneously recorded across the country, 77 stations reported heavy rainfall, with amounts exceeding 200 mm on 28 September.

Among the hydrological gauging stations installed in various rivers across the country, 23 stations recorded water levels surpassing the danger level, and an additional 14 stations recorded water levels exceeding the warning level. The water flow in the Saptakoshi River has reached the highest level in 56 years.

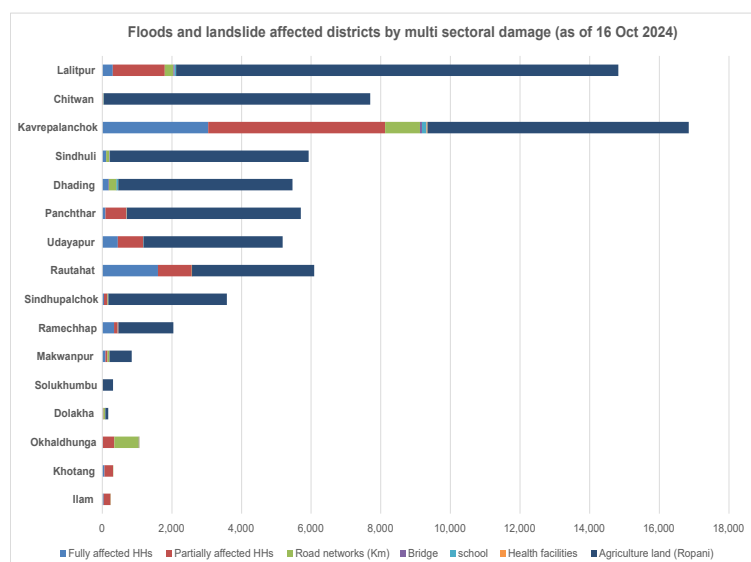


After 22 years, Kathmandu witnessed record rainfall on Friday and Saturday. DHM reported that the rainfall on Saturday was the highest ever in the valley's recorded history, surpassing the previous record set in 2002. Rainfall at the Tribhuvan International Airport station in Kathmandu reached 239.7 millimeters in 24 hours, compared to the previous record of 177 millimeters in 2002.

The continuous heavy rainfall has caused widespread flooding and landslides across Nepal, resulting in 249 fatalities, 18 missing persons, and 177 injured. Rescue operations have saved more than 17000 individuals so far, and efforts to provide shelter, food, and essential supplies are ongoing. Security forces and local bodies are coordinating these relief efforts, with priority given to clearing landslide-blocked roads to facilitate transportation. All three levels of government, security forces, private sectors, development partners, and civil society are working on rescue efforts, emergency relief, response, and the rehabilitation of critical facilities, including schools, health centers, drinking water supplies, and road networks, as well as restoring the affected people back to normal life.

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## Most Affected Districts



## Status of Housing and Settlement

As of 17 Oct 2024, Bagmati Province is the hardest hit, with 4,334 fully affected and 6,593 partially affected households. Koshi Province has 653 fully affected and 2,102 partially affected households, while Madesh Province has 929 fully affected and 3,782 partially affected households.

| Province     | # of fully affected HHs | # of partially affected HHs |
|--------------|-------------------------|-----------------------------|
| Bagmati      | 4,334                   | 6,593                       |
| Koshi        | 653                     | 2,102                       |
| Madesh       | 929                     | 3,782                       |
| <b>Total</b> | <b>5,916</b>            | <b>12,477</b>               |

## Status of Drinking Water

Around **1,639 water supply projects** have been impacted, with extensive damage to critical infrastructure such as intakes, pipelines, water tanks, treatment plants, and household toilets. The total estimated damage is approximately **NPR 5.73 billion**, with major losses including **NPR 630 million** under Kathmandu Upatyaka Khanepani Limited (KUKL) and **NPR 100 million** in the Timal water supply project in Kavre.

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number of Houses Affected       | Approximately 500,000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Number of People Affected       | Approximately 2.5 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Number of Water Supply Projects | Approximately 1,639 (According to information from Federal and Provincial Offices)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Damage Details                  | <ul style="list-style-type: none"> <li>Extensive damage to intakes, sump wells, pipelines, water tanks, treatment plants, and other critical infrastructure.</li> <li>Damage to assets such as GI pipes, HDPE pipes, etc.</li> <li>Damage to structures of wastewater treatment plants.</li> <li>Damage to household and institutional toilets.</li> <li>Under KUKL (Kathmandu Upatyaka Khanepani Limited), approximately NPR 630 million worth of damage to various water supply and sewage management structures.</li> <li>In the Timal water supply project in Kavre alone, damage worth approximately NPR 100 million.</li> </ul> |
| Total Estimated Damage          | Approximately NPR 5.73 billion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Number of Affected Districts    | Kathmandu, Lalitpur, Bhaktapur, Kavre, Sindhupalchok, Ramechhap, Sindhuli, Udayapur, Sunsari, and 58 other districts                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

## Status of Roads and Highways

Multiple highway sections across Koshi, Bagmati, and Lumbini Provinces have been affected due to landslides, floods, and infrastructure damage, leading to road closures. A total of 9 road/highway sections are still affected, including major highways like the **Mechi Highway**, **B.P Highway**, and **Araniko Highway**. Efforts for road clearance and infrastructure repair are ongoing.

| District/Location                     | Name of Road         | Reason for Blockage | Status       |
|---------------------------------------|----------------------|---------------------|--------------|
| Khatahad/Khotehad GP-9, Fokingtar     | Diktel-Gaighat Road  | Landslide           | Road Closed  |
| Dhankuta Benighat GP-7, Gurangse      | Prithvi Highway      | Road Collapse       | One-Way Open |
| Dhankuta Sisakate GP-6, Rishta Bridge | Kalapani Road        | Road Collapse       | One-Way Open |
| Ramche Marshyangdi GP-9               | Prithvi Highway      | Bridge Collapse     | Road Closed  |
| Ramche Khola GP-9, Kalapani           | Ramche-Sindhuli Road | Bridge Collapse     | Road Closed  |

|                           |                   |                |              |
|---------------------------|-------------------|----------------|--------------|
| Kakani GP-1, Piple        | Nuwakot Road      | Landslide      | One-Way Open |
| Kakani GP-1, Kakani Danda | Tribhuvan Highway | Landslide      | One-Way Open |
| Sindhupalchowk GP-1       | Melamchi Highway  | Flood          | Road Closed  |
| Humla Simkot GP-5         | Hilsa Road        | Heavy Snowfall | Road Closed  |

## Key Actions

- Provincial and local governments, along with District Administration Offices, are providing relief materials.
- A total of **NPR 4.32 million** has been distributed as cash assistance to families of deceased individuals, injured persons, and for house damage across Kathmandu, Bhaktapur, Lalitpur, Ramechhap, Sindhuli, Kavrepalanchok, Sindhupalchowk, Dolakha and Dhading districts.
- Bagmati Province has distributed **NPR. 58 million** from the Provincial Disaster Management Fund to support **45 disaster crisis/emergency municipalities**
- Emergency relief materials (food, clothing, and shelter items) have been distributed by local governments, development partners in most affected areas.

## Needs and Priorities

- Temporary shelter for displaced families.
- Socio-technical assistance to enroll affected households into NDRRMA Management Information System for temporary shelter tranche as per the Guidelines for Temporary Housing Construction Grants for Disaster-Affected Households, 2081 (2024).
- Tents with light and other utilities for affected families.
- Emergency WASH facilities that include provisions of safe drinking water, sanitation facilities and hygiene kits.
- Emergency health facilities and disease surveillance to control disease outbreak .
- Provision of food and other humanitarian supplies.
- Construction of temporary learning centers and health facilities.
- Detailed multi- sectoral loss and damage assessments of affected infrastructure.
- Psychosocial Counselling support
- Debris clearance, repair and reconstruction of disrupted road networks and highway.

## Other Needs

- Prefabricated steel truss vehicular bridge – Total of 1050 meters
- Outboard motor with raft – 50 sets
- Heavy lifting drones (for relief material distribution) – 25 Unit
- Combi tools for road traffic accidents (hydraulic) – 60 sets
- Flood light tower (Portable) – 100 units
- Power ascender with rope launcher – 30 sets
- Multi-function transport drone - 16 set



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